

Knowledge and Reasoning

Knowledge

- Knowledge includes facts about the real world entities and the relationship between them
- Characteristics of Knowledge:
 - It is voluminous in nature and requires proper structuring.
 - It may be incomplete and imprecise.
 - It may keep on changing (dynamic).
- There is a need to represent knowledge to ease the development of an intelligent system.
 - Different knowledge representation schemes may be appropriate depending on tasks and circumstances

Motivation

- Intelligent agents/systems are useless without the ability to represent knowledge
- Intelligent system require:
 - **Knowledge** (current state) of the world
 - formally represented using knowledge representation language
 - **Inference mechanisms**: to deduce new facts/knowledge that are unseen by current percepts
 - **Goal**: what it wants to achieve
 - **Effect of its action**: what its own action do in various circumstances

Knowledge-based Agent (KBA)

- Agents can be seen as knowing about their world, and reasoning about their possible courses of action.
- KBA begins with some knowledge of the world and of its actions.
 - It uses logical reasoning to maintain a description of the world as new percepts arrive
 - Deduce a course of actions that will achieve its goals
- One can also design an **autonomous agent** that
 - learns and construct knowledge without human interventions

Components of a KBA

The agent is composed of:

- **Knowledge base**

- It operates by storing a set of sentences (facts and relationship between facts) about the world in its KB.

- **Inference mechanism :**

- With the help of inference mechanism deduce new sentences and use them to decide what action to take

- **Learning mechanism:**

- Adapt to changes in the environment by updating the relevant knowledge

Knowledge Base (KB)

- Contains set of **facts** about the domain expressed in a suitable representation language
 - Each individual representation are called sentences
 - Sentences are expressed in a (formal) **knowledge representation (KR) language**
- A KBA is designed such that there is a way to:-
 - **TELL** it (i.e. to add new sentences to the KB) and **ASK** it (i.e. to query the KBA)
 - When one ASKs a question, the answer should follow from what has been TELLED to the KB previously
 - **Inference mechanism** determines what follows from what has been TELLED to the KB

Knowledge Representation & Reasoning

- **Knowledge Representation (KR):** express knowledge explicitly in a computer-tractable way such that the agent can reason out.
- **Parts of KR language:**
 - **Syntax** of a language: describes the possible configuration to form sentences. **E.g.:** if x & y denote numbers, then $x > y$ is a sentence about numbers
 - **Semantics:** determines the facts in the world to which the sentences refer. **E.g.:** $x > y$ is false when y is greater than x and true otherwise
- **Reasoning:** is the process of constructing new sentences from existing facts in the KB.
 - Proper reasoning ensures that the new configuration represent facts that actually follow from the facts in the KB.

How is Knowledge represented?

- Construct KB by TELLing the agent what it needs to know
 - add one by one the sentences that represent the domain knowledge using **KR language**
- Knowledge is basically represented as “**symbol structures**” (essentially, complex data structures) representing bits of knowledge (objects, concepts, facts, rules, strategies...)
 - Example: “**red**” represents color red.
 - “**mycar**” represents my car.
 - red(mycar)** represents the fact that my car is red.
- Intelligent behavior can be achieved through manipulation of symbol structures
- Build KB following **declarative approach**
 - Rather than procedural language (like C++/Java data structures), use declarative languages (like prolog).

Requirement for KR languages?

- ***Representational adequacy***: the KR language should be effective in representing the knowledge you need for your problem.
 - Is the KR language allows to represent simple or complex knowledge?
- **Understandability**: the representational scheme should enable to do it in a clear, precise and “natural” way.
 - Choose names for symbols that are meaningful.
- **Inferential adequacy**: have appropriate inference methods to reason on knowledge represented and draw new conclusions.
- A balance must be found between effective representation, understandability and Inferential adequacy.

Logic as KR

- A Logic is a **formal language** in which knowledge can be represented such that conclusions can be drawn.
 - It is a declarative language to assert sentences and deduce from sentences.
- **Components of a formal logic** include: syntax, semantics, reasoning and inference mechanism.
 - **Syntax:** what expressions/structures are allowed in the language. Describes how to make sentences
 - E.g. `red(mycar)` is ok, but `mycar(grey or green)` is not.
 - **Semantics:** express what sentences mean, in terms of a mapping to real world.
 - The meaning of a sentence is not *intrinsic* to that sentence. Semantics relate sentences to reality.
 - E.g. `red(mycar)` means that my car is red.
 - **Proof Theory:** how we can draw new conclusions from existing statements in the logic. It is a means of carrying our reasoning using a set of rules.

Why *formal* languages (Logic) ?

- An obvious way of expressing or representing facts and thoughts is by writing them in a natural language such as English, Amharic, etc. However,
 - The meaning of a sentence depends on the sentence itself and on the context on which the sentence was spoken
e.g. Look!
 - Natural languages exhibit **ambiguity**. E.g. small dogs and cats.
 - A single sentence can usually be interpreted in more than one way, possibly inhibiting reasoning. Consider English sentences like:
 - “The boy saw a girl **with a telescope**”
 - “Our shoes are guaranteed to give you **a fit**”
- Ambiguity makes reasoning difficult and incomplete.
 - Hence we need formal languages to express facts and concepts in an unambiguous and well-defined way.

Logical Arguments

- All humans have 2 eyes.
Dawit is a human.
Therefore Dawit has 2 eyes.
- All humans have 4 eyes.
Dawit is a human.
Therefore Dawit has 4 eyes.
- Both are (logically) **valid** arguments. Which statements are **true / false** ?

The other way arguments

- All humans have 2 eyes. Dawit has 2 eyes.
– Therefore Dawit is a human.
- No human has 4 eyes.
– Dawit has 2 eyes.
– Therefore Dawit is not human.
- Both are (logically) **invalid** arguments. Which statements are **true / false** ?

Propositional logic

- A simple language useful for showing key ideas and definitions
- **Syntax:** PL allows facts about the world to be represented as sentences formed from:
 - **Logical constants:** True, False
 - **Proposition symbols ($P, Q, R, \dots$)** are used to represent facts about the world: e.g.: P = "It is hot", Q = "It is humid", R = "It is raining"
 - **Logical connectives:** not ($\neg$), and ($\wedge$), or ($\vee$), implies ($\rightarrow$), is equivalent, if and only if ($\leftrightarrow$).
 - **Parenthesis ():** Used for grouping sentences and to specify order of precedence
 - $(P \rightarrow (Q \rightarrow (\neg(R))))$ vs. $P \rightarrow (Q \rightarrow \neg R)$
 - Precedence order from highest to lowest is: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$
 - The sentence $\neg P \vee Q \wedge R \rightarrow S$ is equivalent to $((\neg P) \vee (Q \wedge R)) \rightarrow S$
 - Binary operators are left associative: so $P \rightarrow Q \rightarrow R$ is $(P \rightarrow Q) \rightarrow R$

Questions:

- Is $(P \vee Q) \vee R$ same as $P \vee (Q \vee R)$?
- Is $(P \wedge Q) \wedge R$ same as $P \wedge (Q \wedge R)$?
- Is $(P \rightarrow Q) \rightarrow R$ same as $P \rightarrow (Q \rightarrow R)$?

Propositional logic (PL)

- A sentence is made by putting symbols together.
 - There are atomic and complex sentences.
- Atomic sentences: consist of propositional symbol (e.g. P, Q, TRUE, FALSE)
 - If S is a sentence, then $\neg S$ is a sentence (not or negation)
 - If S is a sentence, then so is (S)
- Complex sentences are combined by using **connectives** or **parenthesis**:
 - If S and T are sentences, then $S \vee T$, $(S \vee T)$, $(S \wedge T)$, $(S \rightarrow T)$, and $(S \leftrightarrow T)$ are complex sentences, and are called **conjunction**, **disjunction**, **implication**, and **equivalent**, respectively.

Examples: PL sentences for the “weather problem”:

- “If it is hot and humid, then it is raining”: $(P \wedge Q) \rightarrow R$
- “If it is humid, then it is hot” : $Q \rightarrow P$
- “It is humid.”: Q
- **A better way**: Ho=“It is hot”, Hu=“It is humid”, R=“It is raining”

A BNF grammar of sentences in PL

```
S           := <Sentence> ;
<Sentence>  := <AtomicSentence> | <ComplexSentence> ;
<AtomicSentence> := "TRUE" | "FALSE" | "P" | "Q" | "S" ;
<ComplexSentence> := "(" <Sentence> ")" |
    <Sentence> <Connective> <Sentence> | "NOT" <Sentence>;
<Connective> := "NOT" | "AND" | "OR" | "IMPLIES" |
    "EQUIVALENT"
```

Examples: Convert from English to Propositional sentence

Let A = Lectures are active, R = Text is readable, and P = Solomon will pass the exam, then represent the following:

- the lectures are **not** active: $\neg A$
- the lectures are active **and** the text is readable: $A \wedge R$
- either the lectures are active **or** the text is readable: $A \vee R$
- **if** the lectures are active, **then** the text is not readable: $A \rightarrow \neg R$
- the lectures are active **if and only if** the text is readable: $A \leftrightarrow R$
- if the lectures are active, then if the text is not readable, Solomon will not pass the exam: $A \rightarrow (\neg R \rightarrow \neg P)$

Terminology

- **Valid sentence:** A sentence is **valid sentence** or **tautology** if and only if it is True under all possible interpretations in all possible worlds.

Example: “It’s raining or it’s not raining.” $(R \vee \neg R)$.

- **Satisfiable:** A sentence is satisfiable if and only if there is some interpretations in some world for which the sentence is True.

Example: “It is raining or it is humid”. $R \vee Q$, R

- **Unsatisfiable:** A sentence is unsatisfiable (inconsistent sentence or self-contradiction) if and only if it is not satisfiable, i.e. a sentence that is False under all interpretations. The world is never like what it describes.

Example: “It’s raining and it's not raining.” $R \wedge \neg R$

Terminology

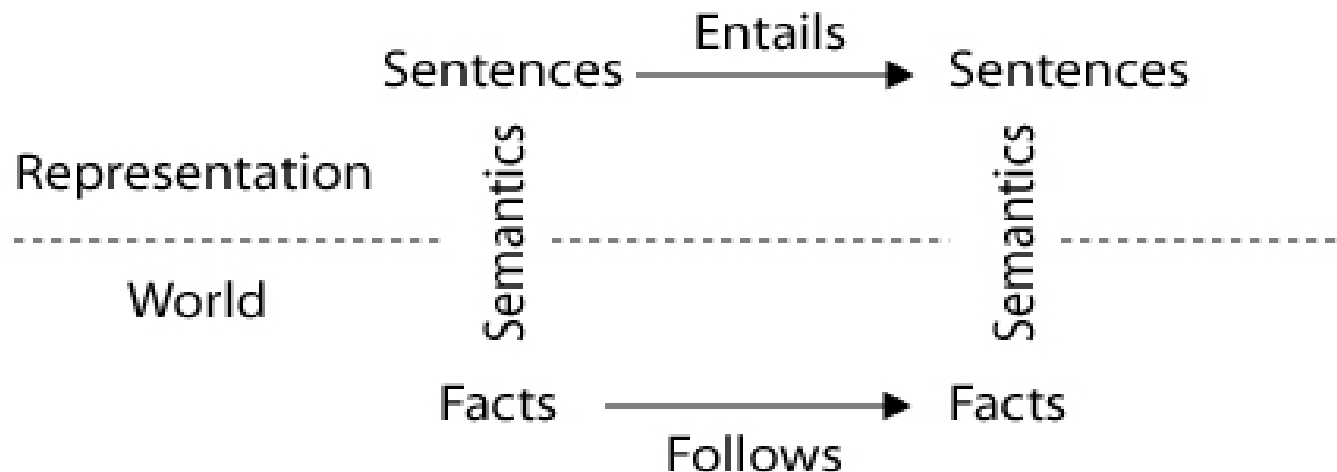
- **Entailment:** mirrors the relation of one fact following from another.
 - New sentences are generated that are necessarily true, given that the old sentences are true.
 - KB entails α ($KB \models \alpha$) if and only if the conclusion α is true in every logically possible worlds in which all the premises in KB are true.
- **Derivation:**

KB $\vdash$ Q, Q is derived from KB if there is a proof consisting of a sequence of valid inference steps starting from the premises in KB and resulting in Q

Semantics

- **Facts and their representation:**

- Facts are part of the world, where as their representation must be encoded in some way that can be physically stored within an agent
- A semantics maps sentences to facts in the world. In the simplest case, the semantics determines what objects in the world are referenced by which objects in the language.
- The property of one fact following from another should be mirrored by the property of one sentence being logically *entailed* by another.



Semantics

- Specify the interpretation of the proposition symbols & constants, and the meanings of the logical connectives
 - **Truth Tables**: define the semantics of sentences. The following table shows truth table for the five logical connectives.

Note: P and Q can be **any** sentence, including complex sentences.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
True	True	False	True	True	True	True
True	False	False	False	True	False	False
False	True	True	False	True	True	False
False	False	True	False	False	True	True

- Sentences in propositional logic tell you about what is true or false.
 - $P \wedge Q$ means that both P and Q are true.
 - $P \vee Q$ means that either P or Q is true (or both)
 - $P \rightarrow Q$ means that if P is true, so is Q .

Semantics (Complex Sentences)

- Complex sentences such as $(R \wedge S) \rightarrow (\neg R \vee S)$ are defined by a process of **decomposition**.
 - First determine the meaning of $(R \wedge S)$ and of $(\neg R \vee S)$.
 - Then combine them using the definition of the $\rightarrow$ function

R	S	$\neg R$	$R \wedge S$	$\neg R \vee S$	$(R \wedge S) \rightarrow (\neg R \vee S)$
True	True	False	True	True	True
True	False	False	False	False	True
False	True	True	False	True	True
False	False	True	False	True	True

- Can you write the truth table for:
 $\neg(P \rightarrow Q) \rightarrow (R \rightarrow \neg P)$

Logical equivalence

- Two sentences are logical equivalent iff they are true in same models. $\alpha \equiv \beta$ if and only if $\alpha \models \beta$ and $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{De Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{De Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Inference Rules

- **Inference** is used to create new sentences that logically follow from a given set of sentences in the KB.
 - It captures patterns of inferences that occur over & over again.
 - Once a rule is established, it can be used to make inferences without going through the tedious process of building truth tables
- An inference rule $\{\alpha, \beta\} \vdash \Theta$ consists of two sentence patterns α & β called the conditions (premises) and one sentence pattern Θ called the conclusion
 - If α & β match two sentences of KB, then the corresponding Θ can be inferred according to the rule
- Given set of inference rules (I) and set of sentences (KB); Inference is the process of applying successive inference rules from I to KB, each rule inferring new facts and adding its conclusion to KB
 - **Example: Modus Ponens:** $\{\alpha \Rightarrow \beta, \alpha\} \vdash \beta$

Inference Rules

RULE	PREMISE	CONCLUSION
Modus Ponens	$A, A \rightarrow B$	B
Modus Tolens	$\neg B, A \rightarrow B$	$\neg A$
And Elimination	$A \wedge B$	A
And Introduction	A, B	$A \wedge B$
Or Introduction	A	$A_1 \vee A_2 \vee \dots \vee A_n$
Double Negation Elimination	$\neg \neg A$	A
Unit Resolution	$A \vee B, \neg B$	A
Resolution	$A \vee B, \neg B \vee C$	$A \vee C$
Hypothetical Syllogism	$P \rightarrow Q, Q \rightarrow R$	$P \rightarrow R$

- In the case of modus ponens, if A is true and $A \rightarrow B$ is true, then conclude B is true.

Two important properties of inference

Completeness: If $KB \models Q$ then $KB \vdash Q$

- A logic is **complete** if it is capable of proving all consequences that can be represented in it.
 - If Q is entailed by a set of sentences KB , then Q can be derived from KB using inference rules.
 - Hence, inference produces all entailments, or all valid sentences can be proved from the premises.
- A set of inference rules is **complete** if every entailed sentences can be obtained by applying some finite succession of these rules
 - Modus ponens *alone* is not complete,
e.g.: from $A \Rightarrow B$ and $\neg B$, we cannot get $\neg A$

Soundness

Soundness: If $KB \vdash Q$ then $KB \models Q$

- A logic is **sound** if it preserves truth (i.e. if a set of premises are all true, any conclusion drawn from those premises must also be true).
 - If Q is derived from a set of sentences KB using a given set of rules of inference, then Q is entailed by KB .
 - Hence, inference produces only real entailments, or any sentence that follows deductively from the premises is valid.
 - We can proof soundness by constructing the truth table.
- An inference rule is **sound** if it generates only entailed sentences
- All inference rules previously given are sound
 - A rule is sound if its conclusion is true whenever the premise is true.
E.g. modus ponens: $\{\alpha \Rightarrow \beta, \alpha\} \vdash \beta$
- The following rule: $\{\alpha \Rightarrow \beta, \beta\} \vdash \alpha$ is unsound, which does not mean $\vdash$ it is useless

Formal Proofs

- A **proof** is a sequence of steps followed to generate sentences, where each sentence is
 - either a premise or a sentence/formula derived from earlier sentences (of the proof) using some of the rules of inference.
 - The last step of the proof should deduce the formula (also called goal or query) that we want to prove.

Example: The “**weather problem**”. Proof whether it is raining or not.

1. Q	Premise	“It is humid”
2. $Q \rightarrow P$	Premise	“If it is humid, it is hot”
3. $(P \wedge Q) \rightarrow R$ raining”	Premise	“If it’s hot & humid, it’s
4. P	Modus Ponens(1,2)	“It is hot”
5. $P \wedge Q$	And Introduction(1,4)	“It is hot and humid”
6. R	Modus Ponens(3,5)	“It is raining”

Example 1

Construct formal proof of validity for the following problem:

If the investigation continues, then new evidence is brought to light. If new evidence is brought to light, then several leading citizens are implicated. If several leading citizens are implicated, then the newspapers stop publicizing the case. If continuation of the investigation implies that the newspapers stop publicizing the case, then the bringing to light of new evidence implies that the investigation continues. The investigation does not continue. Therefore, new evidence is not brought to light.

Represent using PL and proof the conclusion that
“new evidence is not brought to light”.

Solution

Let

C: The investigation continues.

B: New evidence is brought to light.

I: Several leading citizens are implicated.

S: The newspapers stop publicizing the case.

1. $C \rightarrow B$

2. $B \rightarrow I$

3. $I \rightarrow S$

4. $(C \rightarrow S) \rightarrow (B \rightarrow C)$

5. $\neg C$

6. $C \rightarrow I$ 1,2 (*Hypothetical Syllogism*)

7. $C \rightarrow S$ 6,3 (*Hypothetical Syllogism*)

8. $B \rightarrow C$ 7,4 (*Modus Ponens*)

9. $\neg B$ 8,5 (*Modus Tollens*)

Example 2

Nathan is either intelligent or a good actor. If Nathan is intelligent, then he can count from 1 to 10. Nathan can only count from 1 to 2. Therefore Nathan is a good actor.

Represent using PL and proof the conclusion that Nathan is a good actor or not.

Let:

I: Nathan is intelligent

A: Nathan is a good actor.

C: Nathan can count from 1 to 10.

More examples

- Represent the following problem and infer the third sentence from the first two:

Every person is mortal.

Abebe is a person.

Therefore, Abebe is mortal.

- In PL use propositional symbols to stand for **all or part of** each sentence. Hence,

– P = “person”; M = “mortal”; A = “Abebe”

– So the above sentences are represented as:

$P \rightarrow M$, $A \rightarrow P$ (Hypothesis). We can infer: $A \rightarrow M$

- Although the third sentence is entailed by the first two, we need an explicit symbol, A , to represent an individual, Abebe, who is a member of the classes “person” and “mortal.”

– To represent other individuals we must introduce separate symbols for each one, which means for representing the fact that all individuals who are “people” are also “mortal.”

Propositional logic is a weak language

- PL cannot handle even a domain with small worlds. The problem is that there are just too many propositions to handle since it only has one representational device: the **proposition**
- In PL world consists of just facts. It is hard to :
 - Identify individuals: E.g., Mary, 3
 - Describe properties of (or relations between) individuals. E.g. Belete is taller than Abreham
 - Generalize for a given universe. E.g., all triangles have 3 sides
- First-Order Logic (FOL) is expressive enough to concisely represent this kind of situation.
 - FOL represents objects and relations between objects, variables, and quantifiers in addition to propositions
 - Every elephant is gray:* $\forall x (\text{elephant}(x) \rightarrow \text{gray}(x))$
 - There is a white alligator:* $\exists x (\text{alligator}(X) \wedge \text{white}(X))$
- Prove that “my dog Fido is Nice, given that “all dogs are Nice.”
 - This requires to get at the structure and meanings of statements (where FOL is useful).

Conclusion

- Due to the **ambiguity** in natural languages there is a need to specify knowledge through the use of **formal** languages
 - Formal languages give us a way to remove ambiguity and provide methods for **automating inference**. **PL** is a first move in this direction
- **Inference**: The process of deriving new sentences from old one
 - **Sound** inference: derives true conclusions given true premises.
 - **Complete** inference: derives all true conclusions from a set of premises.
- A **valid sentence** is true in all worlds under all interpretations.
 - If an implication sentence can be shown to be valid, then given its premise - its consequent can be derived.
- Different logics make different **commitments** about what the world is made of and what kind of beliefs we can have regarding the facts.
 - Logics are useful for the commitments they do not make because lack of commitment gives the knowledge base engineer more freedom.
- **PL** commits only to the existence of facts that may or may not be the case in the world being represented.
 - Using a simple syntax and semantic, it illustrates the process of inference.
 - PL quickly becomes impractical, even for very small worlds.